Hao LIU

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Research profile: computational environmental-data researcher with experience in urban pollution, source inversion, and sparse observations under meteorological variability. Research interests include physics-based atmospheric modelling of pollutant sources, transport, and chemical processing across local to regional scales, using machine learning where it adds practical value.

Education

Imperial College London — Sep 2025 - Sep 2026

MSc Environmental Data Science and Machine Learning *Merit*

- Training includes inversion and optimization, advanced programming, cloud computing, and environmental data science.

Guizhou University — Sep 2021 - Jun 2025

BEng Information Management and Information Systems

- Core training in algorithms, object-oriented programming, linear algebra, statistics, data visualization, machine learning, and data mining.

Publications and Manuscripts

Liu, H. (2026). **RE-BAE v1.0: a Bayesian approximation-error method for reliability-aware sparse urban emission inversion. Submitted to *Geoscientific Model Development* (GMD).** Preprint: https://doi.org/10.21203/rs.3.rs-10542170

Liu, H. **Physics-Regularized Neural Networks for Urban Air Quality Prediction Under Meteorological Variability: Emission Source Estimation from Sparse Observations in London.** *Theoretical and Applied Climatology*. **Revised manuscript resubmitted after first-round review; awaiting second-round decision.** Preprint: researchsquare.com/article/rs-8810489/v1

Liu, H., & Qin, X. (2024). **Target Detection of Safety Protective Gear Using the Improved YOLOv5.** DOI: doi.org/10.1109/CAIT64506.2024.10962947

Research Experience

Urban Emission Inversion · Independent Research — 2025 - Present

Physics-Guided Source Estimation under Meteorological Variability

- Formulated urban emission-source estimation as an inverse problem, using sparse surface observations to infer latent source parameters under changing meteorological conditions.
- Developed a physics-regularized neural modelling workflow that embeds physical consistency into learning rather than relying on unconstrained black-box prediction.
- Developed RE-BAE, a Bayesian approximation-error method that builds a structured representativeness-error covariance from fine-minus-coarse wind differences, and evaluated posterior calibration (coverage, CRPS, sharpness) with real winds over Los Angeles and Dallas–Fort Worth.
- Examined how monitoring-network density interacts with correlated representativeness error, showing that when this correlation is ignored, added monitors narrow posterior intervals without improving accuracy, so that coverage of nominal 90% intervals falls below 20%.
- Applied sensitivity analysis across observation constraints to assess the robustness of source estimates and inform monitoring-network design.

Research Experience (continued)

Imperial College London · Independent Research Project **Jun 2026 - Sep 2026**

Separating Rainfall-Associated $PM_{2.5}$ Reductions from Normal Temporal Change: A Five-Year Matched-Control Study in Tropical Singapore *Supervisors: Dr Yves Plancherel; Dr Fangxin Fang*

- Analysed five years of hourly rainfall, $PM_{2.5}$, and meteorological data across Singapore's five reporting regions to separate rainfall-associated removal from normal temporal change.
- Matched rainfall events to rain-free controls by region, month, local onset hour, weekday/weekend type, event duration, and pre-rain $PM_{2.5}$; 1,436 of 2,116 high-$PM_{2.5}$ events obtained an eligible matched control (67.9% matching rate).
- Estimated an excess reduction of 2.49 µg/m³ (95% CI: 2.00-2.98), with a conservative adjusted estimate of 2.08 µg/m³ (95% CI: 1.49-2.67); effects increased with pre-rain $PM_{2.5}$, rainfall duration, and mean intensity.
- Showed that prediction intervals for individual events crossed zero, supporting inference on the mean effect across many events rather than deterministic claims for a single rainfall event. Thesis repository

Leadership & Community Engagement

Book Donation Initiative · Event Organiser **2022**

- Coordinated on-site operations for a university book donation drive, including venue setup, donor registration, queue management, and volunteer shift handovers; collected approximately 200 books and arranged their transfer to the Volunteer Association's External Relations team.

Technical Skills

Programming	Python, C/C++, Java, SQL
Scientific computing	NumPy, pandas, finite-difference PDE solvers, Bayesian inversion, uncertainty quantification, sensitive analysis
Modelling & ML	Physics-guided modelling, time-series modelling, PyTorch, TensorFlow, scikit-learn, YOLOv5
Research tools	Git, LaTeX, Jupyter Notebook